PyGameMaker — Tutorial 7: Platformer

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Platformer: Run, Jump, Collect** (4 pages, about 25-30 minutes). Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase. If something does not match, look at **Stuck?** before you call your teacher.

What You Will Make

A platform game: your character falls with gravity, runs and jumps between platforms, collects coins, avoids spikes (each hit costs a life) and reaches a flag to win.

Phase 1: A Jumping Player

- ☐ **1.** Create two sprites (32×32): spr_player and spr_ground.
- ☐ **2.** Create obj_ground (sprite spr_ground, **Solid**) and obj_player (sprite spr_player).
- ☐ **3.** In obj_player, **Create** event: *Set gravity* direction **270**, strength **0.5**.
- ☐ **4.** Add **Keyboard: Left Arrow (held)** with *Set horizontal speed to -4* and **Right Arrow (held)** with *4*.
- ☐ **5.** Add **Keyboard: No key** with *Set horizontal speed to 0*. Do **not** use *Stop movement* here: it would also cancel gravity.
- ☐ **6.** Add **Key press: Up Arrow** with *Set vertical speed to -10*.
- ☐ **7.** Add **Collision with obj_ground** with *Stop movement*.
- ☐ **8.** Create room_level1 (800×480, light blue background, Snap to Grid 32×32). Place ground along the bottom, platforms in the air, and the player standing on the ground on the left. **Keep the space above the player free** so it can jump.

> **You should see:** press **F5**. The player falls onto the ground, runs left and right, and the up arrow makes it jump about three tiles high. It lands on platforms.

Phase 2: Coins and Hazards

- ☐ **9.** Create sprites spr_coin, spr_spike and spr_flag and objects obj_coin, obj_spike and obj_flag (no events on them).
- ☐ **10.** In obj_player add **Collision with obj_coin**: *Add to score 10*, then *Destroy other instance*.
- ☐ **11.** Add **Collision with obj_spike**: *Restart room*.
- ☐ **12.** Add **Collision with obj_flag**: *Show message* "You Win!".
- ☐ **13.** In the room, put coins on platforms, spikes near pits, and the flag at the right end.

> **You should see:** press **F5**. Coins vanish and are worth 10. A spike restarts the level. The flag shows "You Win!".

Phase 3: Game Controller

- [ ] **14.** Create obj_game_controller (no sprite).
- [ ] **15. Game Start** event (from *Other Events*): *Set score to 0* and *Set lives to 3*.
- [ ] **16. Draw** event: *Draw score at x 10, y 10* and *Draw lives at x 200, y 10*.
- [ ] **17.** In obj_player's spike collision, add *Add to lives -1* **before** *Restart room*.
- [ ] **18.** Place obj_game_controller anywhere in the room.

You should see: press **F5**. Score and lives are shown. Each spike costs one life and restarts the room, and the lives stay lower.

Game Start, not Create. A Create event runs again every time the room restarts. If the lives were set to 3 in Create, every spike would give them back and you would never run out.

Stuck?

What is wrong	Most likely cause	What to do
The player falls through the ground	obj_ground is not Solid, or the player has no **Collision with obj_ground**	Tick **Solid**; add the event with *Stop movement*
The player does not fall at all	The **Set gravity** action is missing from **Create**, or **No key** uses *Stop movement*	Add gravity; change **No key** to *Set horizontal speed to 0*
The player will not jump	The Up event is *held* instead of *Key press*, or something solid is right above the player	Use **Key press: Up Arrow**; keep the space above free
The player jumps again and again while holding Up	It is a Keyboard (held) event	Use **Key press** so each press is one jump
Coins do nothing	The coin event is on the coin instead of the player, or missing	Put **Collision with obj_coin** on obj_player
Spikes never cost lives	The lives are set in **Create** instead of **Game Start**, or *Add to lives -1* is missing	Move the actions to **Game Start**; add the action before *Restart room*
The score and lives do not show	obj_game_controller is not in the room	Place it in the room
The flag does nothing	No **Collision with obj_flag** on the player	Add it

Challenges

- **Try this (5 minutes):** change the gravity (0.5) or the jump speed (-10) and see how the jump feels.
- **Push further:** add a moving platform or a patrolling enemy; make the flag go to the next room.
- **Invent:** add a double jump, or a Game Over message when the lives reach 0.

Vocabulary

Term	What it means
Gravity	A force that pulls an object down (direction 270) every step
Vertical speed	How fast something moves up (negative) or down (positive)
Key press	An event that happens once when a key goes down
Hazard	Something that hurts the player, like a spike
Game Start	An event that runs once, when the whole game begins

Check Yourself

Write your answers in the notes below.

1. Why do we use *Set horizontal speed to 0* and not *Stop movement* for **No key**?
2. Why is the jump speed negative?
3. Why are the score and lives set in **Game Start** and not in **Create**?

My Notes